Data Science and Artificial Intelligence

Machine Learning

Nearest Neighbour

Lecture No. 3

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Recap of Previous Lecture

- Topic
- Topic
- Topic
- Topic
- Topic

KNN'S

→ Problem

→ distance

→ Algo

Topics to be Covered

- Topic — Questions
- Topic — • time and space Complexity
- Topic — • How KNN solve problem of missing data
- Topic
- Topic

Stay Strong
BE
Positive

REVISION

Problems in KNN → Curse of dimensionality

* need extensive Computation/ Calculation

1. dimension inc → feature selection (PCA)

2. Range of dimension inc → (Normalisation)

Cross Validation is done to find best K for data
↳ hyperparameter.

Confusion matrix Can be used for calculating accuracy of model in classification (On Test set)
↳ Accuracy.

Mean absolute error $\Rightarrow \frac{1}{\text{No of test point}} \sum_{i=1}^{T} |y_i - \hat{y}_i|$ in Regression.

accuracy on Test data

2 class.

Pred. \ Actual	1	0
1	TP	FP
0	FN	TN

3 class

Pred \ Actual	1	2	3
1	T_1	F_1	F_1
2	F_2	T_2	F_2
3	F_3	F_3	T_3

$$\Rightarrow \text{Accuracy} = \frac{T_1 + T_2 + T_3}{\text{Total}}$$

Nearest Neighbour Method

Nearest Neighbour method of Supervised Learning

Problem in K-NN Method

1. Increase in number of features
2. Increase in range of features

What is the curse of dimensionality...

Nearest Neighbour Method

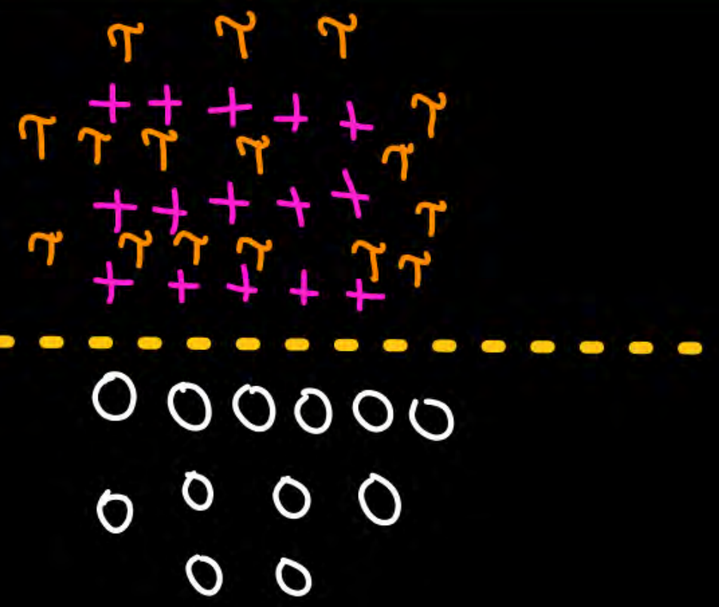
Nearest Neighbour method of Supervised Learning

Some Questions

(3 points) Is it possible for a 2-class 1-NN classifier to always classify all new examples as positive even though there are negative examples in the training data? If yes, show an example. If no, briefly explain.

- 2class 1NN,
- Is it possible that all test point → +ve Class

Yes



Nearest Neighbour Method

Nearest Neighbour method of Supervised Learning

Bias and Variance trade off in K-NN

So the general trend for bias and variance on K is ...

done

Nearest Neighbour Method

Nearest Neighbour method of Supervised Learning

K should not be multiple of number of classes

K=odd

- If we have 3 classes in categorical data
  $\Rightarrow K \neq 3, 6, 9 ---$

- $K \neq$ multiple of No of Classes

Let K=3, 3NN

we may get 3NN with Class 1,2,3

K=6, 6NN

1,1,2,2,3,3

Nearest Neighbour Method

Nearest Neighbour method of Supervised Learning

Hamming distance

If the x values or dimensions are categoriqal in nature

Rain	Climate	Humidity
Yes	Hot	Humid
No	Cold	not Humid

Categorical nature

The euclidian and Manhattan are calculated for points with dimensions of Continuous/Real number.

Rain	Climate	Humidity
1	0	1
0	1	1
1	1	1

← data Point

1 0 1
0 1 1

Hamming distance

Hamming dist ⇒ No of locations where the bit values differ

2

Same as $\sum_{l=1}^{D} \left| x_i^l - x_j^l \right|$

1 1 0
0 0 1

3

If Rain has 4 Categories

Low Rain → 00

No Rain → 01

Heavy Rain → 10

Thunderstorm → 11

Rain	Climate	Humity
00	0	0
01	1	1
10	1	1
11	0	1

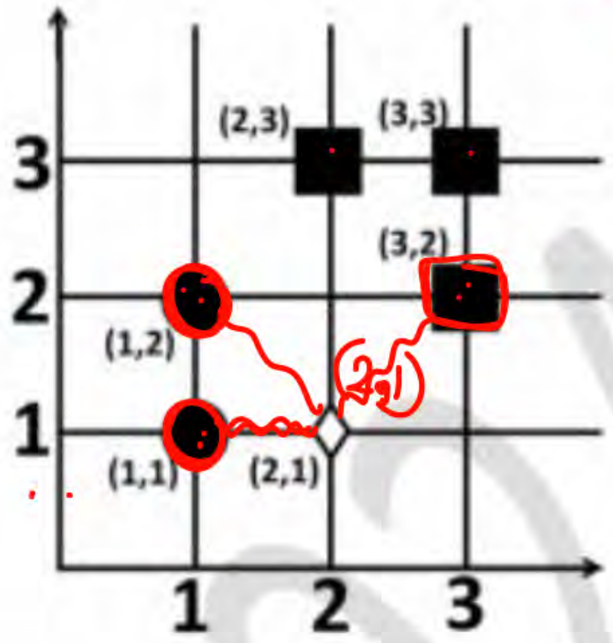
{ 01 1 0
10 1 0 }

2 Hamming dist.

Nearest Neighbour Method

Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k-nearest neighbor algorithm for which the diamond (◇) shaped data point is assigned the label square is ___5.___.

Q.



K=1 ⇒ Circle

K=3 ⇒ 2C, 1Sq
↳ Circle

K=5 ⇒ 3Sq, 2C
↳ Sq

1. What is K-Nearest Neighbor (K-NN)?

a. A supervised learning algorithm

b. An unsupervised learning algorithm

c. A reinforcement learning algorithm

d. A clustering algorithm

2. In K-NN, what does "K" represent?

a. The number of clusters

b. The number of neighbors to consider

c. The number of features

d. The number of iterations

3. Which distance metric is commonly used in K-NN for calculating distances between data points?

a. Euclidean distance

b. Pearson correlation

c. Hamming distance

d. Manhattan distance

4. In K-NN, how is the class label of a data point determined when K is an odd number?

a. By a majority vote of the K nearest neighbors

b. By averaging the class labels of the K nearest neighbors

c. By selecting the class label of the nearest neighbor

d. By randomly selecting a class label

5. What happens when K is too small in K-NN?

a. The model becomes overly complex

b. The model becomes overly simple

c. The model may overfit the data

d. The model may underfit the data

6. In K-NN, what is the impact of increasing K (the number of neighbors)?

a. The model's bias decreases

b. The model's variance decreases

c. The decision boundary becomes more complex

d. The decision boundary becomes smoother

Kinc
Biasinc
Vardec

d

1) [True or False] k-NN algorithm does more computation on test time rather than train time.

A) TRUE

B) FALSE

2) In the image below, which would be the best value for k assuming that the algorithm you are using is k-Nearest Neighbor.

A) 3

B) 10

C) 20

D 50

3) Which of the following distance metric can not be used in k-NN?

A) Manhattan

B) Minkowski

C) Tanimoto

D) Jaccard

E) Mahalanobis

F) All can be used

4) Which of the following option is true about k-NN algorithm?

A) It can be used for classification

B) It can be used for regression

C) It can be used in both classification and regression

5) Which of the following statement is true about k-NN algorithm?

1. k-NN performs much better if all of the data have the same scale
2. k-NN works well with a small number of input variables (p), but struggles when the number of inputs is very large
3. k-NN makes no assumptions about the functional form of the problem being solved

a

A) 1 and 2

B) 1 and 3

C) Only 1

D) All of the above

6) Which of the following machine learning algorithm can be used for imputing missing values of both categorical and continuous variables?

A) K-NN

B) Linear Regression

C) Logistic Regression

1st	2nd	3rd	Label
5	6	3	10
3	5	6	15
2	3	3	20
5	2	1	6
6	6	9	8

we use 1st and 3rd dimension

as X

3.6	5	3
3.16	2	3
5.38	5	1
4.24	6	9

(3, 6)

3NN $\Rightarrow \frac{6+3+6}{3}$

$\Rightarrow 5$

7) Which of the following is true about Manhattan distance?

A) It can be used for continuous variables

B) It can be used for categorical variables

C) It can be used for categorical as well as continuous

D) None of these

8) Which of the following distance measure do we use in case of categorical variables in k-NN?

1. Hamming Distance
2. Euclidean Distance
3. Manhattan Distance

A) 1
B) 2
C) 3
D) 1 and 2
E) 2 and 3
F) 1,2 and 3

10) Which of the following will be Manhattan Distance between the two data point A(1,3) and B(2,3)?

$|1-2| + |3-3| = 1$

Suppose, you have given the following data where x and y are the 2 input variables and Class is the dependent variable.

11) Suppose, you want to predict the class of new data point x=1 and y=1 using eucludian distance in 3-NN. In which class this data point belong to? → class 1.

12) In the previous question, you are now want use 7-NN instead of 3-KNN which of the following x=1 and y=1 will belong to? → -ve ✓

	x	y	Class
1)	-1	1	−
2)	0	1	+
3)	0	2	−
4)	1	-1	−
5)	1	0	+
6)	1	2	+
7)	2	2	−
8)	2	3	+

1, 1 →

1) $\sqrt{(1-(-1))^2+(1-1)^2} \Rightarrow 2$

2) $\sqrt{(1-0)^2+(1-1)^2} \Rightarrow 1$

3) $\sqrt{(1-0)^2+(1-2)^2} \Rightarrow \sqrt{2}$

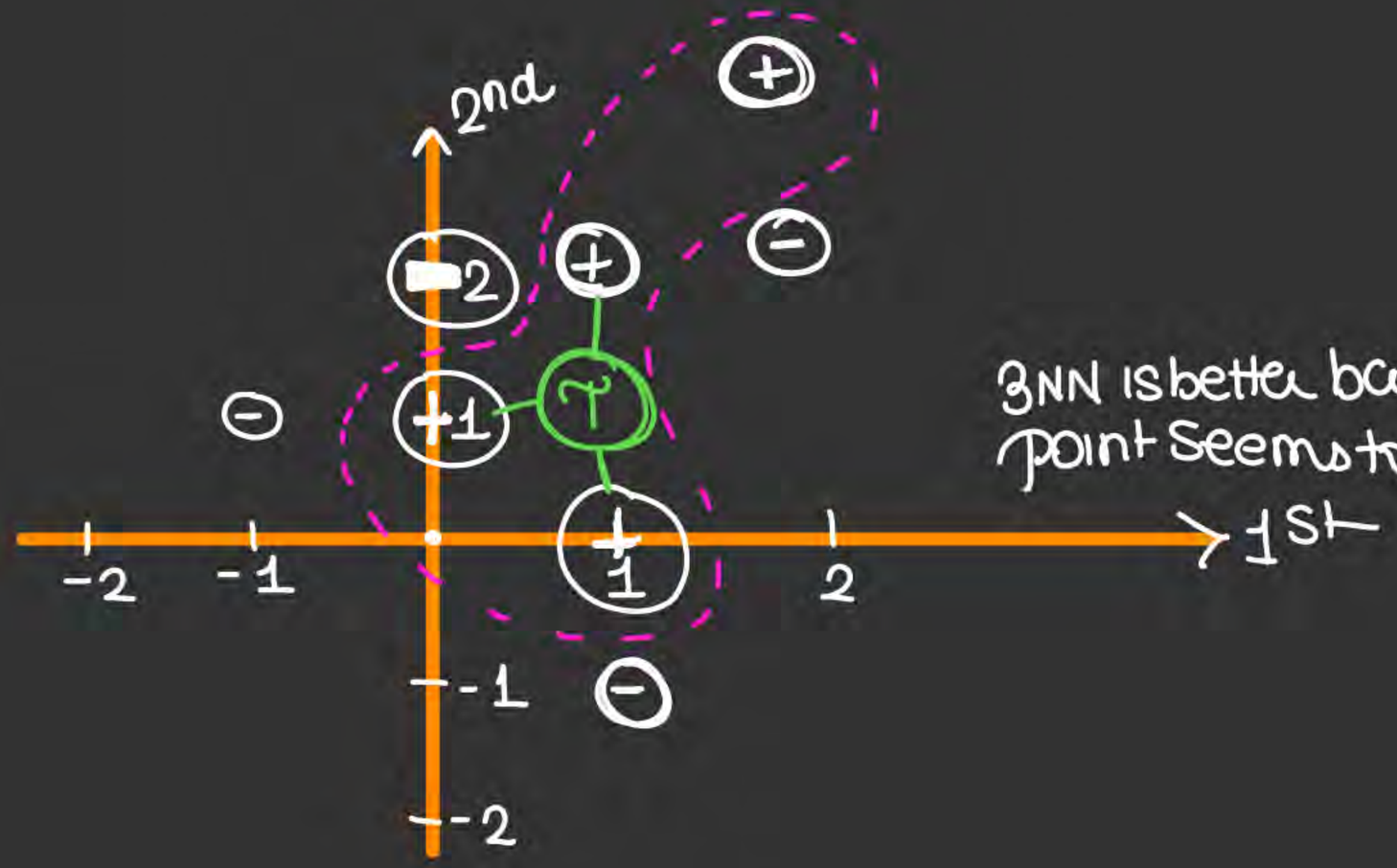
4) $\sqrt{(1-1)^2+(1-(-1))^2} \Rightarrow 2$

5) $\sqrt{(1-1)^2+(1-0)^2} = 1$

6) $\sqrt{(1-1)^2+(1-2)^2} = 1$

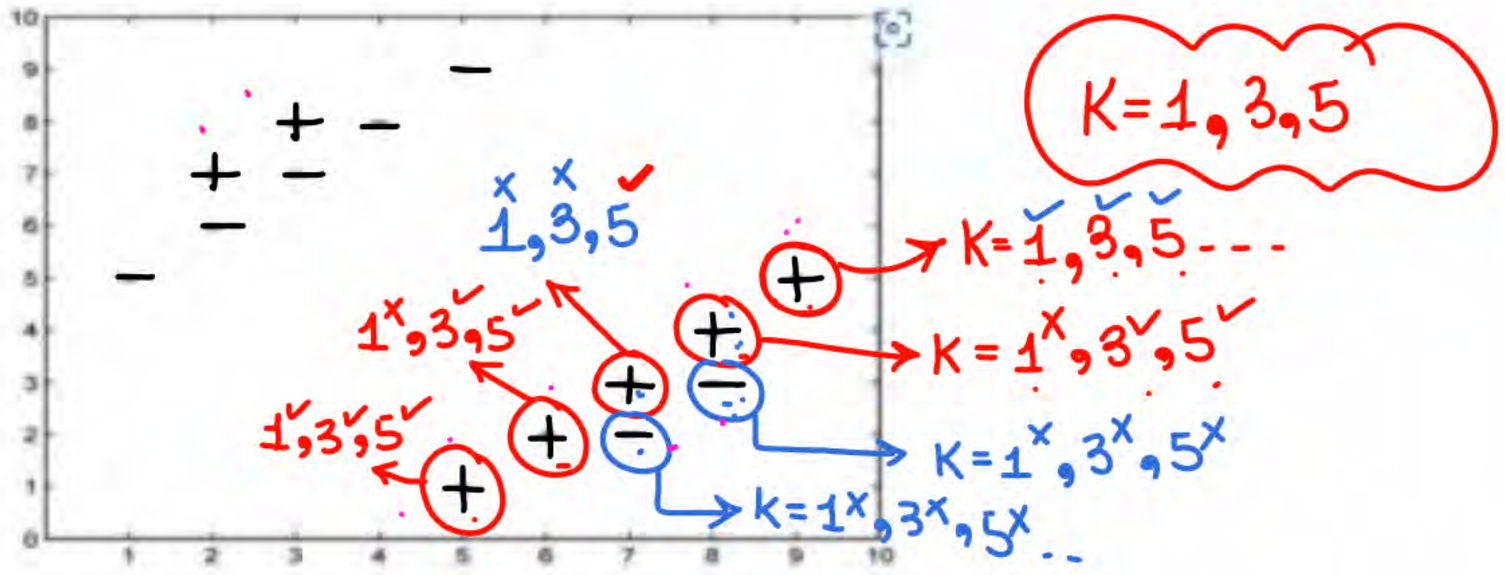
7) $\sqrt{(2-1)^2+(2-1)^2} = \sqrt{2}$

8) $\sqrt{(2-1)^2+(3-1)^2} = \sqrt{5}$



3NN is better bcoz Test point seems to be in ⊕ zone

Suppose you have given the following 2-class data where "+" represent a postive class and "" is represent negative class.



K datapoints

doocv

13 Points Training	1 Validation

K=1 ✗

K=3 ✗

K=5 ✓

13) Which of the following value of k in k-NN would minimize the leave one out cross validation accuracy?

15) Which of the following will be true about k in k-NN in terms of Bias?

A) When you increase the k the bias will be increases

B) When you decrease the k the bias will be increases

C) Can't say

D) None of these

19) In k-NN it is very likely to overfit due to the curse of dimensionality. Which of the following option would you consider to handle such problem?

1. Dimensionality Reduction

2. Feature selection

A) 1

B) 2

C) 1 and 2

D) None of these

20) Below are two statements given. Which of the following will be true both statements?

1. k-NN is a memory-based approach is that the classifier immediately adapts as we collect new training data.
2. The computational complexity for classifying new samples grows linearly with the number of samples in the training dataset in the worst-case scenario.

C

A) 1

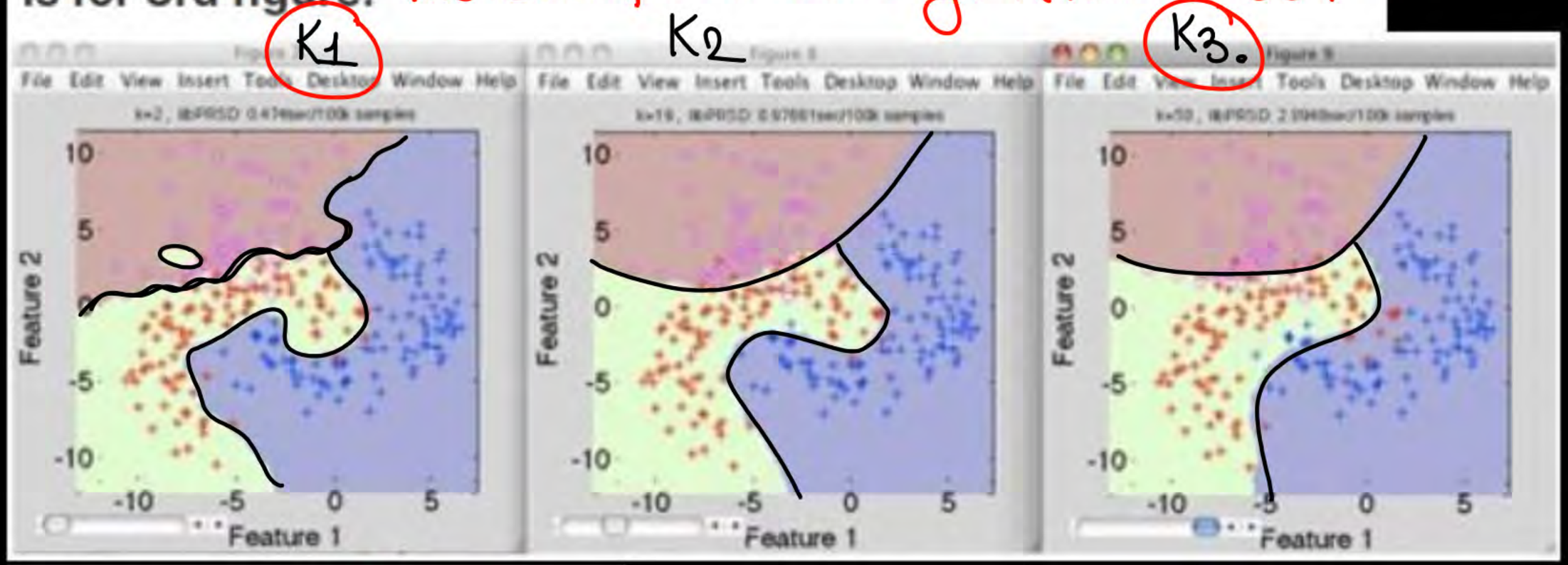
B) 2

C) 1 and 2

D) None of these

21) Suppose you have given the following images(1 left, 2 middle and 3 right), Now your task is to find out the value of k in k-NN in each image where k1 is for 1st, k2 is for 2nd and k3 is for 3rd figure.

more complex boundary. K value small



A) k1 > k2> k3

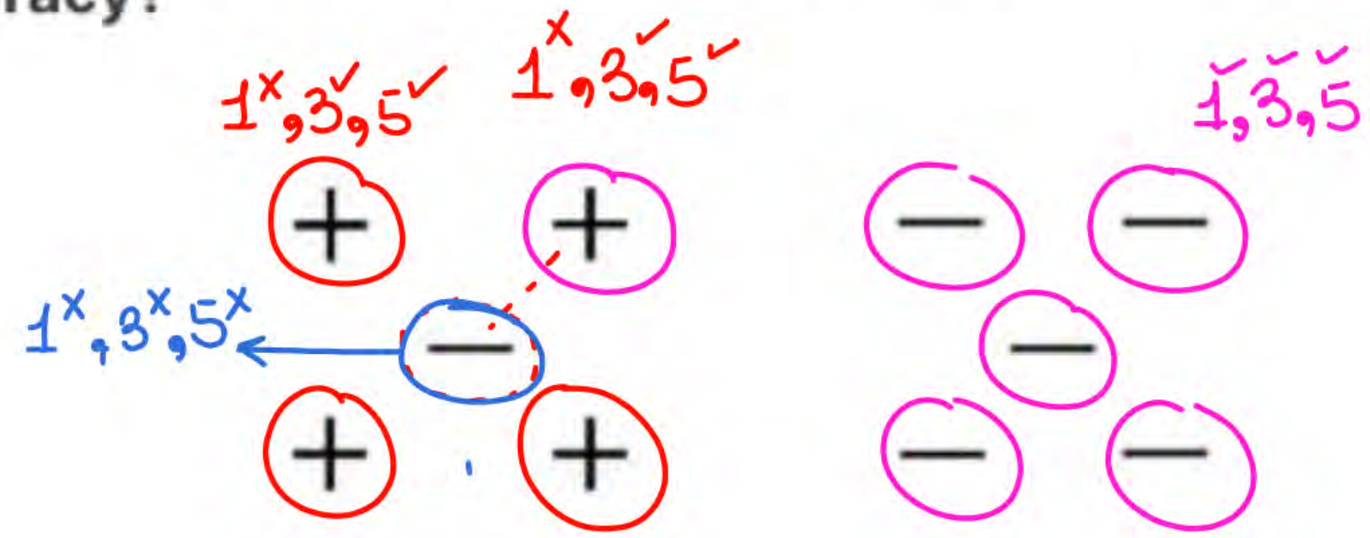
B) k1<k2

C) k1 = k2 = k3

D) None of these

$K_1 < K_2 < K_3$

22) Which of the following value of k in the following graph would you give least leave one out cross validation accuracy?



A) 1

B) 2

C) 3

D) 5

28) Following are the two statements given for k-NN algorthm, which of the statement(s)

is/are true?

C

1. We can choose optimal value of k with the help of cross validation

2. Euclidean distance treats each feature as equally important

A) 1

B) 2

C) 1 and 2

D) None of these

4. We have data from a survey and objective testing with two attributes A and B to classify whether a special paper tissue is good or not. Here are four training samples given in the table. Now the factory produces a new paper tissue that pass laboratory test with A = 3 and B = 7. If K = 3, then 'Good' is the classification of this new tissue?

H.W

A	B	C = Classification
7	6	Bad
7	4	Bad
4	4	Good
2	4	Good

a) True
b) False

2. In a regression task using K-NN, if K is set to 4, and the four nearest neighbors have target values of 8, 10, 12, and 14, what is the predicted value for the data point?

a. 10

b. 11

c. 12

d. 13

y_p = avg of y's

$= \frac{8+10+12+14}{4} \Rightarrow 11$

6. In a K-NN regression task with K set to 3, if the three nearest neighbors have target values of 5, 7, and 9, and their distances from the data point are 2.0, 2.5, and 3.0, what is the predicted value using inverse distance weighting?

a. 5.67
b. 7.33
c. 8.00
d. 9.00

generally $y_p = (\text{avg } y)$

but we can give weights to the y values

So if y values are y_i, weight is w_i

y ⇒	w
5,	1/2
7,	1/2.5
9,	1/3

$$y_p = \frac{\sum y_i w_i}{\sum w_i}$$

$$\frac{5/2 + 7/2.5 + 9/3}{1/2 + 1/2.5 + 1/3} = 6.729$$

8. In a K-NN regression task with K set to 4, if the four nearest neighbors have target values of 10, 12, 15, and 18, and their distances from the data point are 2.0, 2.5, 3.0, and 3.5, what is the predicted value using the inverse distance squared weighting method?

a. 13.04

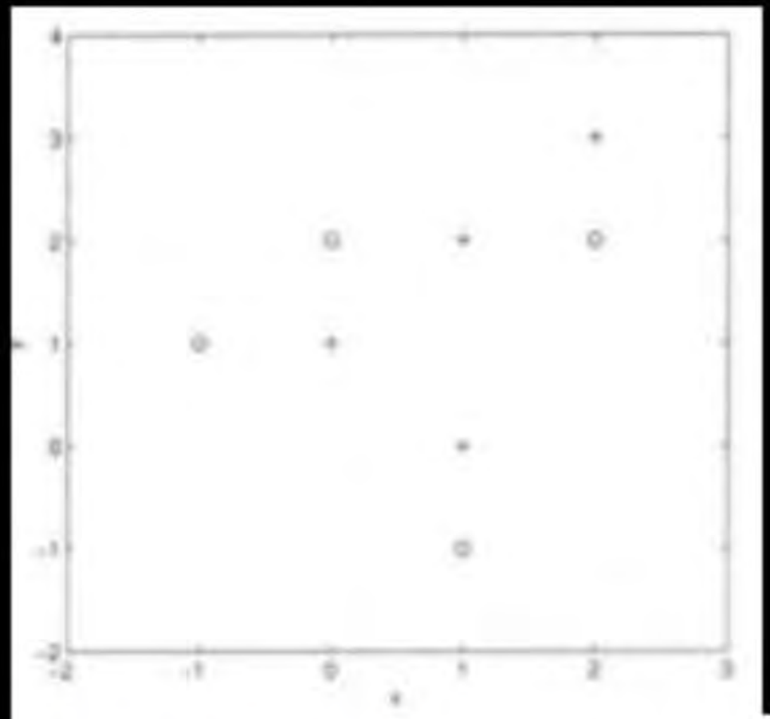
b. 14.47

c. 15.23

d. 16.21

$$\Rightarrow \left\{ \begin{array}{ll} 10 & 1/2^2 \\ 12 & 1/2.5^2 \\ 15 & 1/3^2 \\ 18 & 1/3.5^2 \end{array} \right\}$$

$$y_P = 12.53.$$



11) Suppose, you want to predict the class of new data point x=1 and y=1 using eucludian distance in 3-NN. In which class this data point belong to?

THANK - YOU